Algorithms

Lecture # 27

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Fractional Knapsack Problem

Fractional Knapsack Problem

- There are n items; the i^{th} item is worth v_i and weighs w_i.
- Items can either be put in the knapsack or not.
- The goal was to maximize the value of items without exceeding the total weight limit of W.

Fractional Knapsack Problem

Fractional Knapsack Problem

- In contrast, in the fractional knapsack problem, the setup is exactly the same.
- But, one is allowed to take fraction of an item for a fraction of the weight and fraction of value.

Fractional Knapsack Problem

Fractional Knapsack Problem

- The 0-1 knapsack problem is hard to solve.
- However, there is a simple and efficient greedy algorithm for the fractional knapsack problem.

Fractional Knapsack Problem

Fractional Knapsack Problem

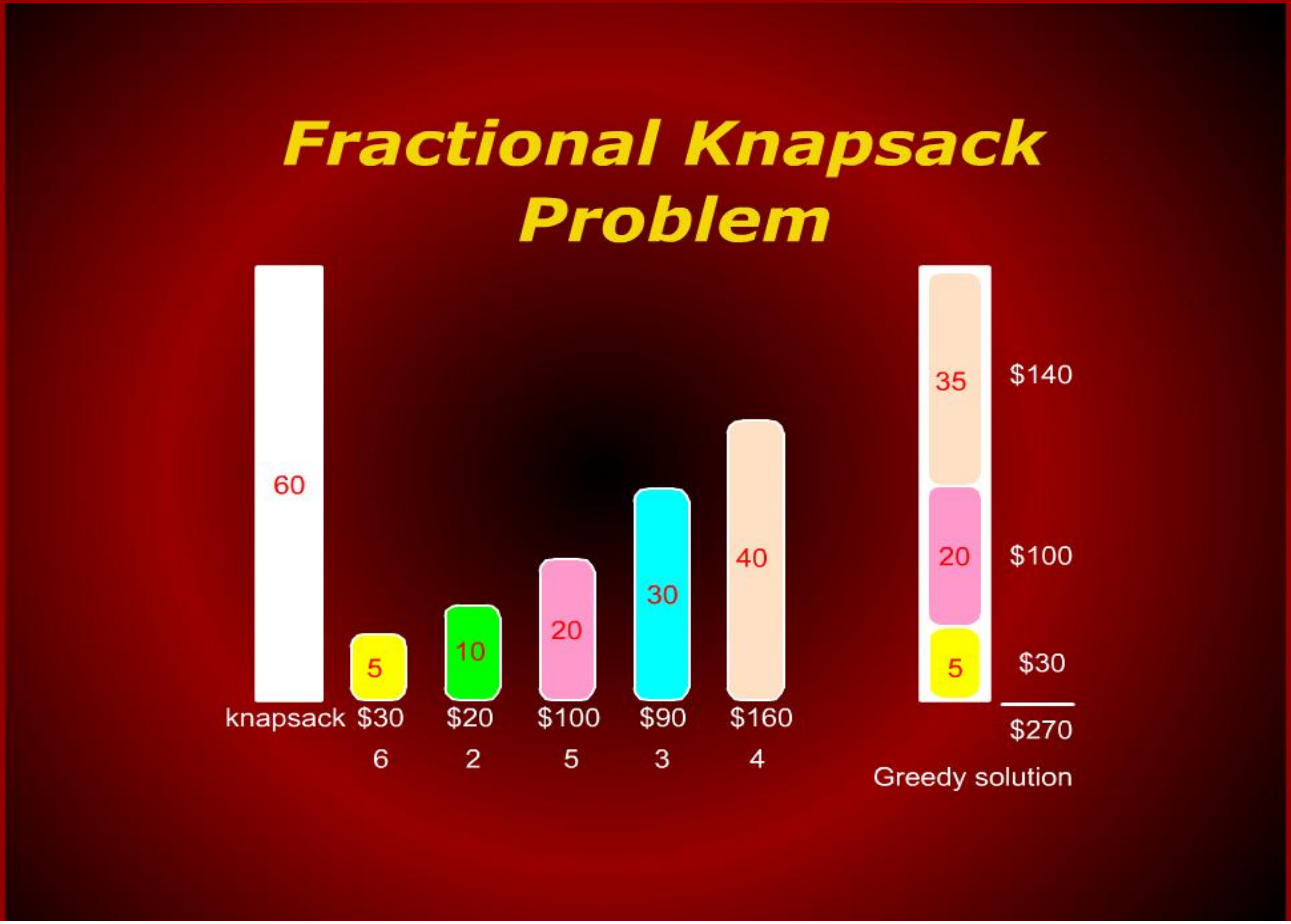
- let $\rho_i = v_i/w_i$ denote the *value per unit weight* ratio for item i.
- Sort the items in decreasing order of ρ_i.
- Add items in decreasing order of ρi.
- If the item fits, we take it all.

Fractional Knapsack Problem

Fractional Knapsack Problem

- At some point there is an item that does not fit in the remaining space.
- We take as much of this item as possible thus filling the knapsack completely.

Fractional Knapsack Problem



Fractional Knapsack Problem

Fractional Knapsack Problem

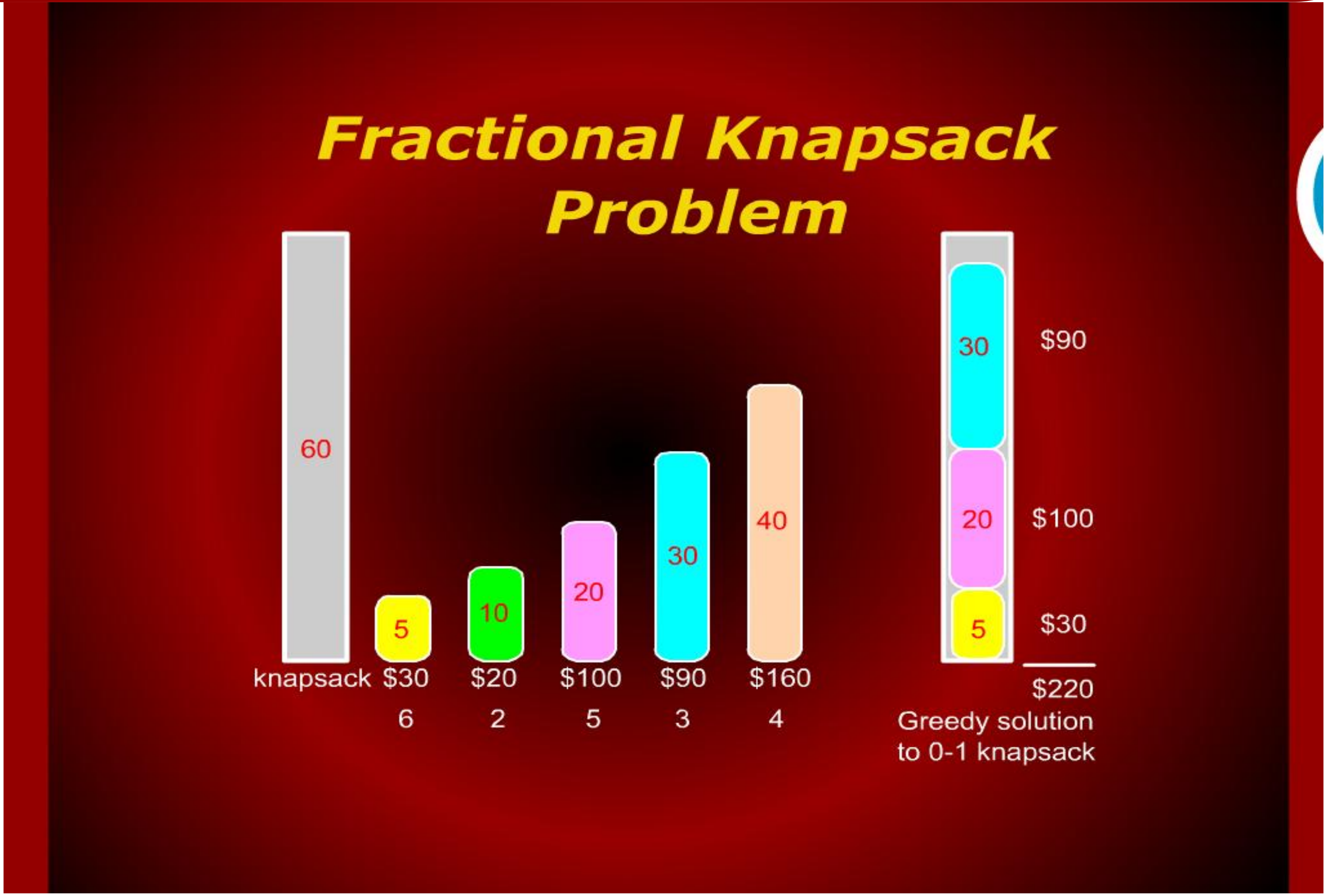
- It is easy to see that the greedy algorithm is optimal for the fractional knapsack problem.
- Given a room with sacks of gold,silver and bronze, one (thief?) would probably take as much gold as possible.

Fractional Knapsack Problem

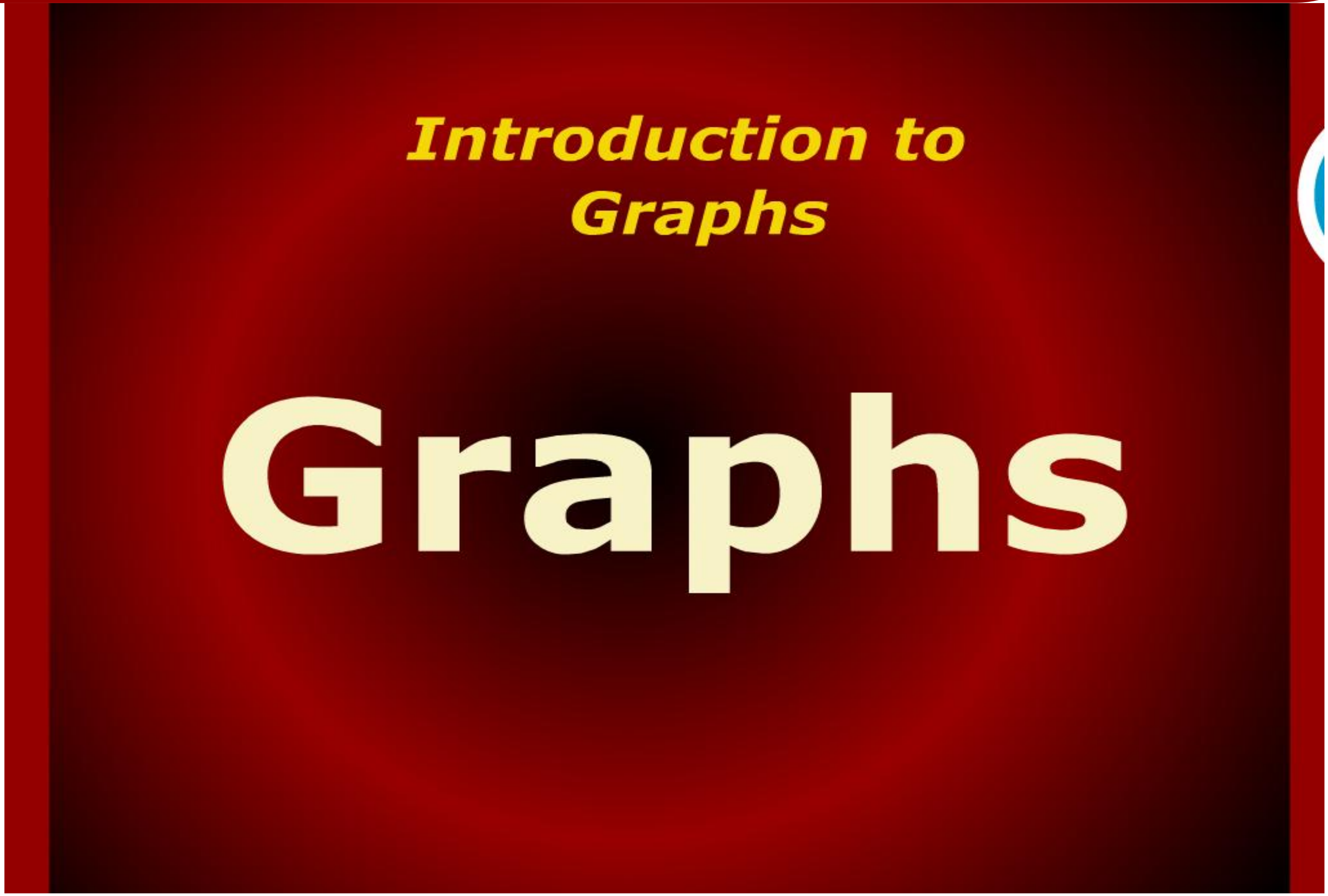
Fractional Knapsack Problem

- We can also observe that the greedy algorithm is not optimal for the 0-1 knapsack problem.

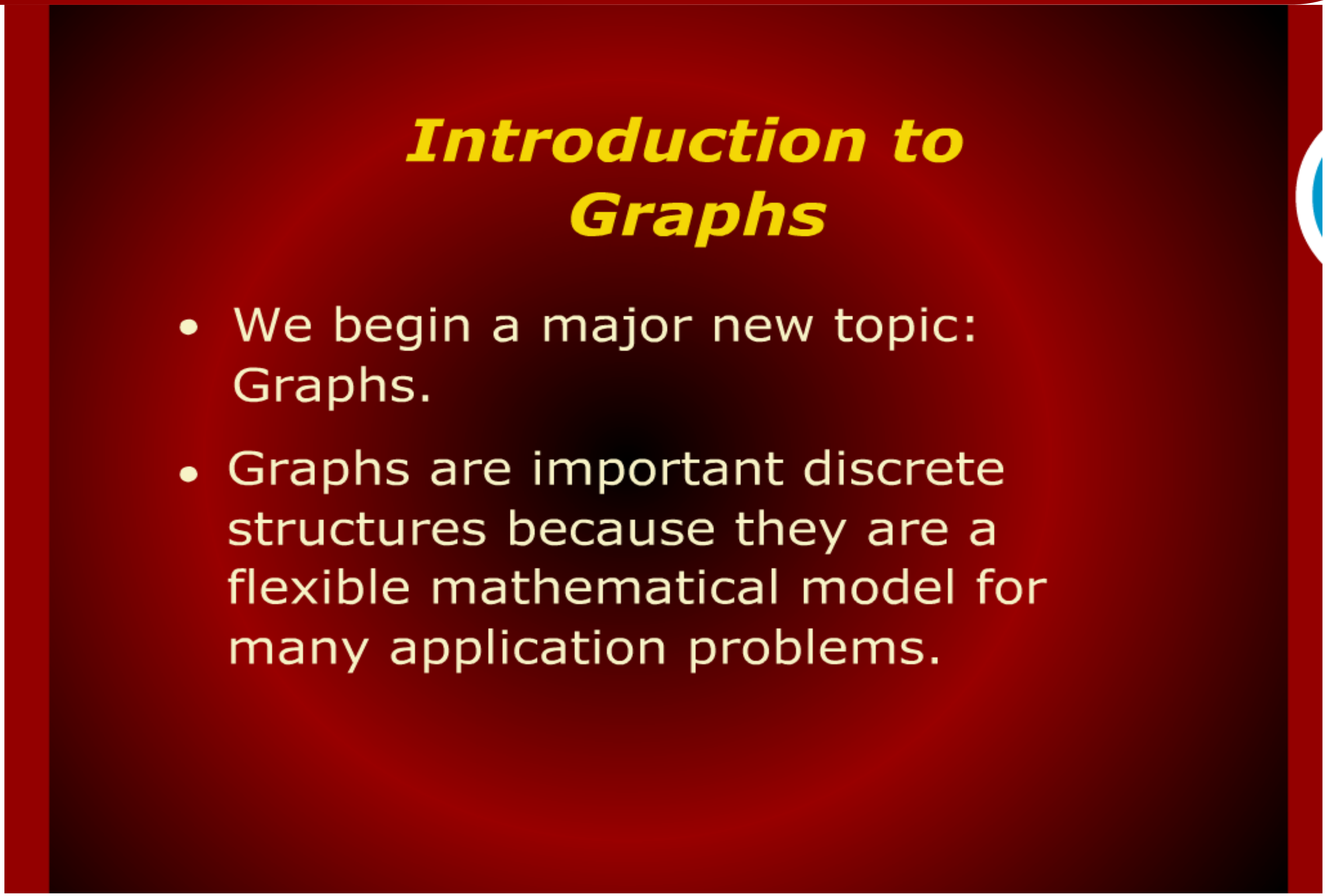
Fractional Knapsack Problem



Introduction to Graphs



Introduction to Graphs



Introduction to Graphs

Introduction to Graphs

Any time there is

- a set of objects and
- there is some sort of “connection” or “relationship” or “interaction” between pairs of objects,

a graph is a good way to model this.

Introduction to Graphs

Introduction to Graphs

Examples:

- computer and communication networks
- transportation networks, e.g., roads
- VLSI, logic circuits
- surface meshes for shape description in computer-aided desigr and GIS

Introduction to Graphs

Introduction to Graphs

- transportation networks, e.g., roads
- VLSI, logic circuits
- surface meshes for shape description in computer-aided desigr and GIS
- precedence constraints in scheduling systems.

Graphs and Digraphs

Graphs and Digraphs

A graph G = (V, E) consists of

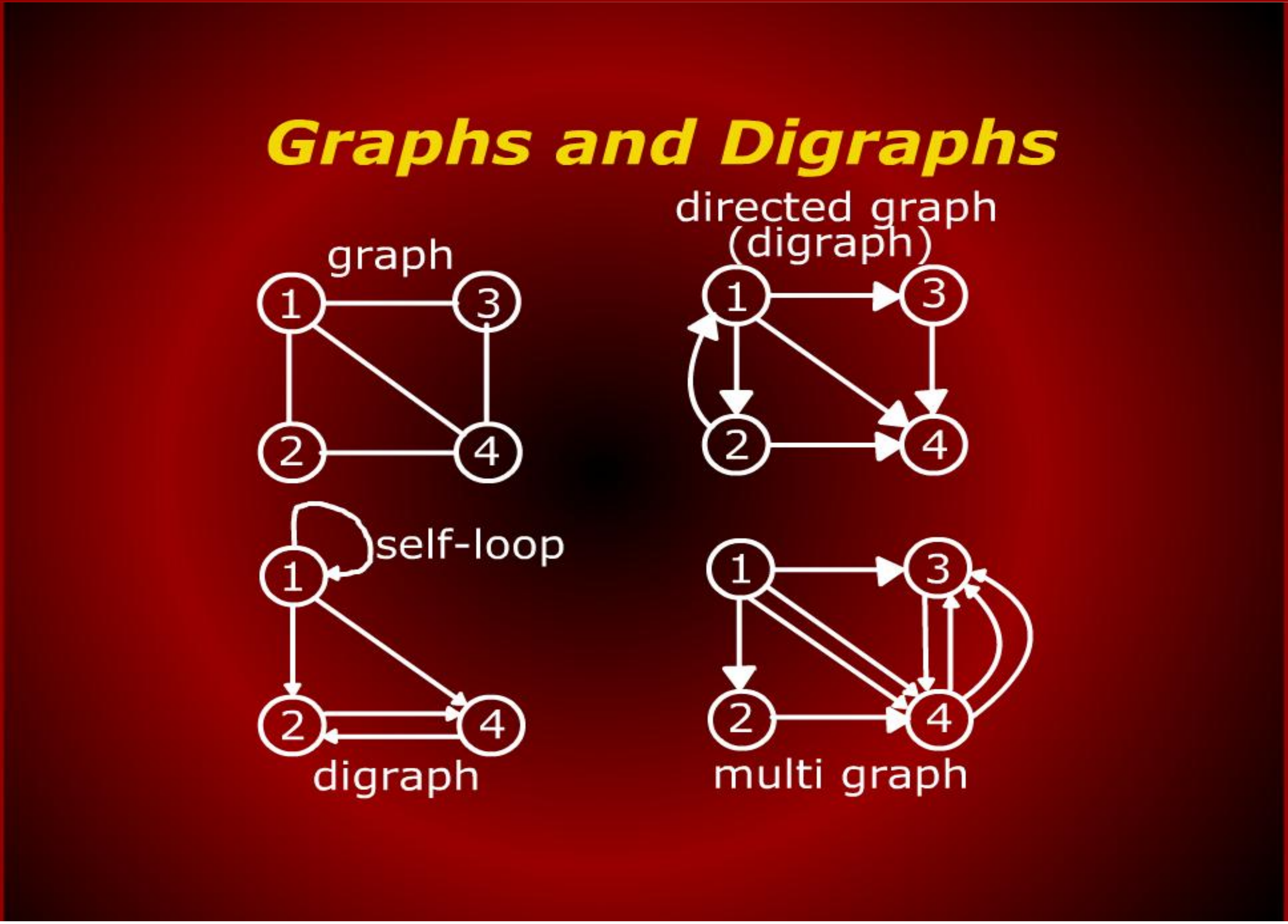
- a finite set of vertices V (or nodes) and
- E, a binary relation on V called edges.

Graphs and Digraphs

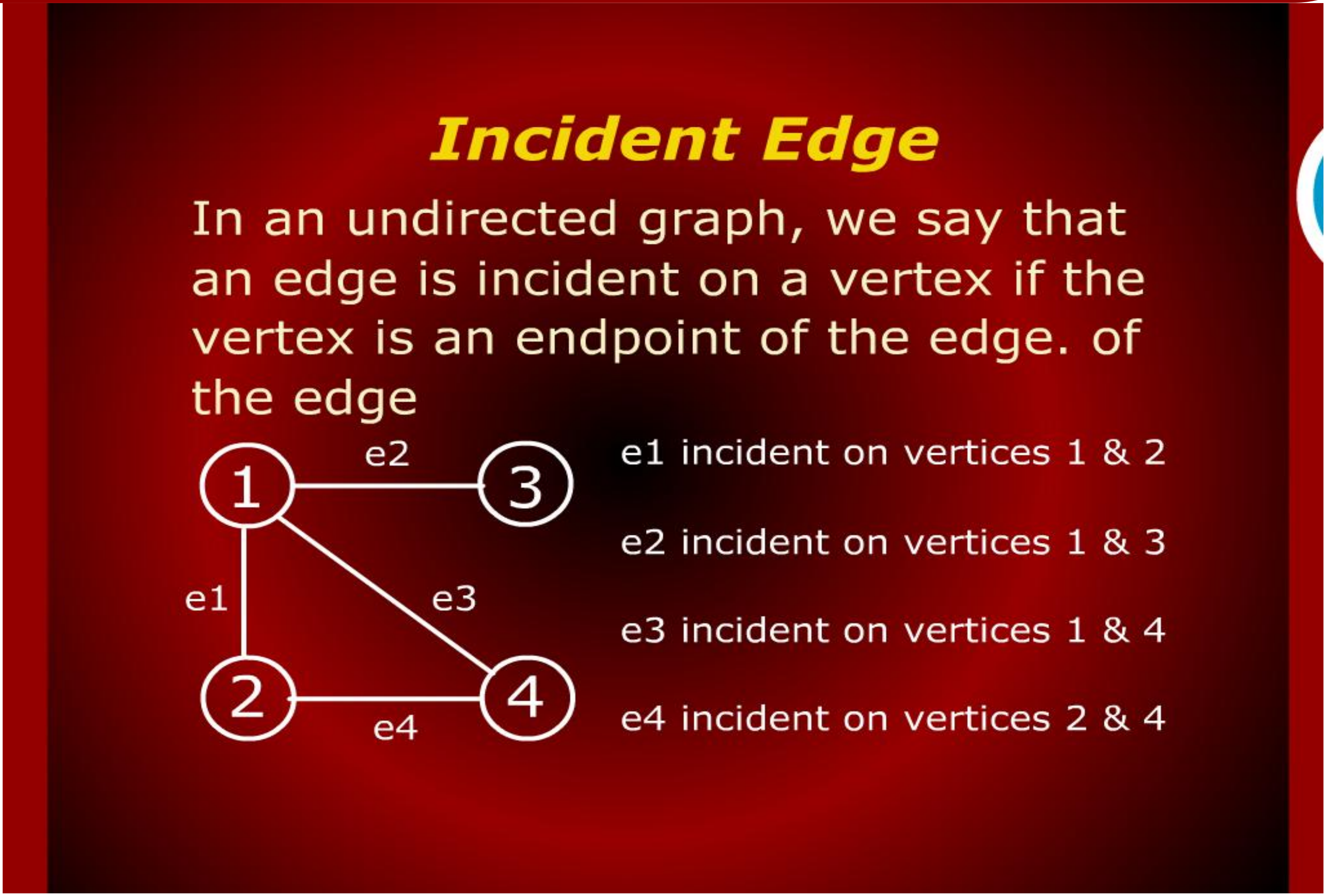
Graphs and Digraphs

- E is a set of pairs from V.
- If a pair is *ordered*, we have a *directed* graph.
- For *unordered* pair, we have an *undirected* graph.

Graphs and Digraphs



Incident Edge

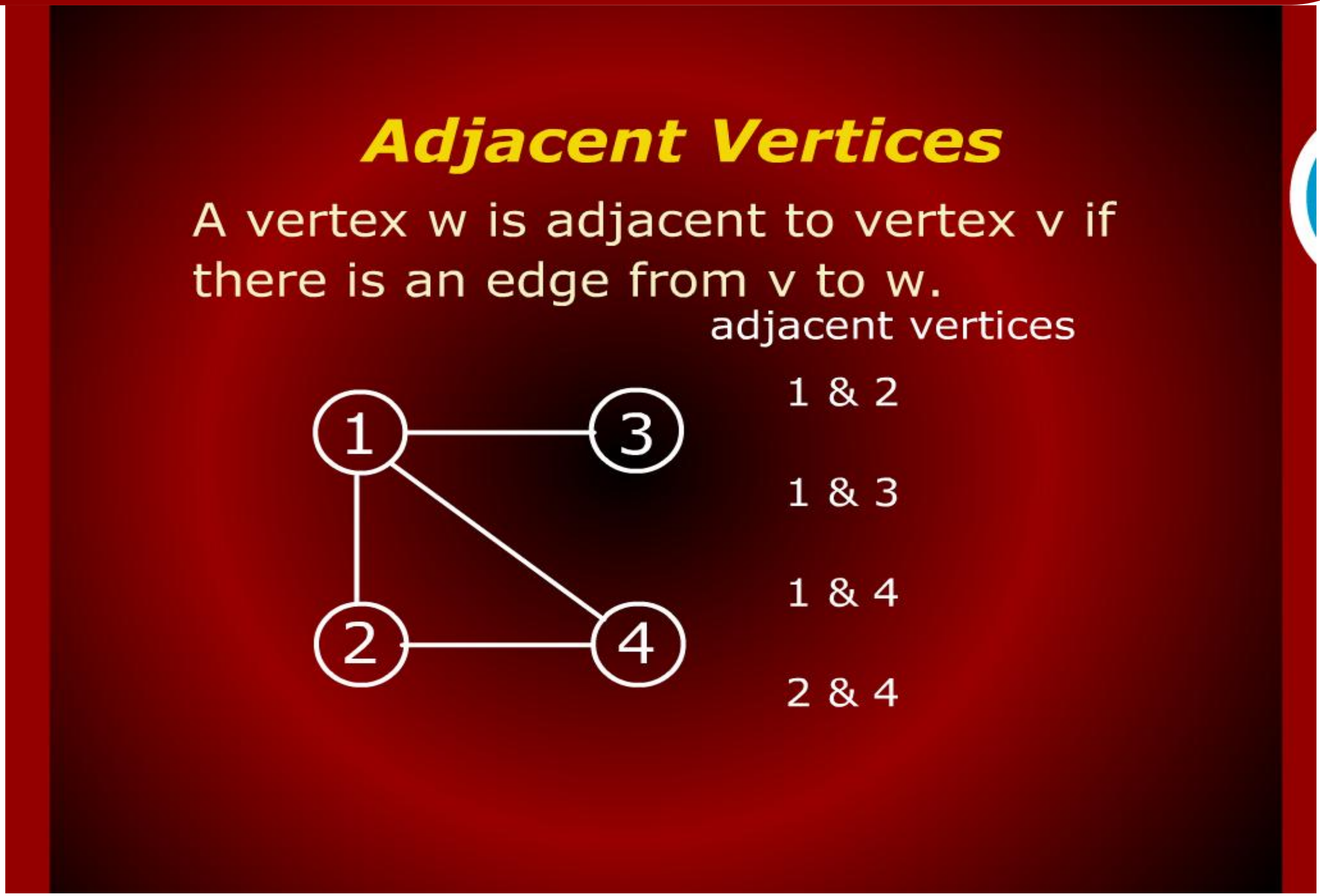


Degree of a Vertex

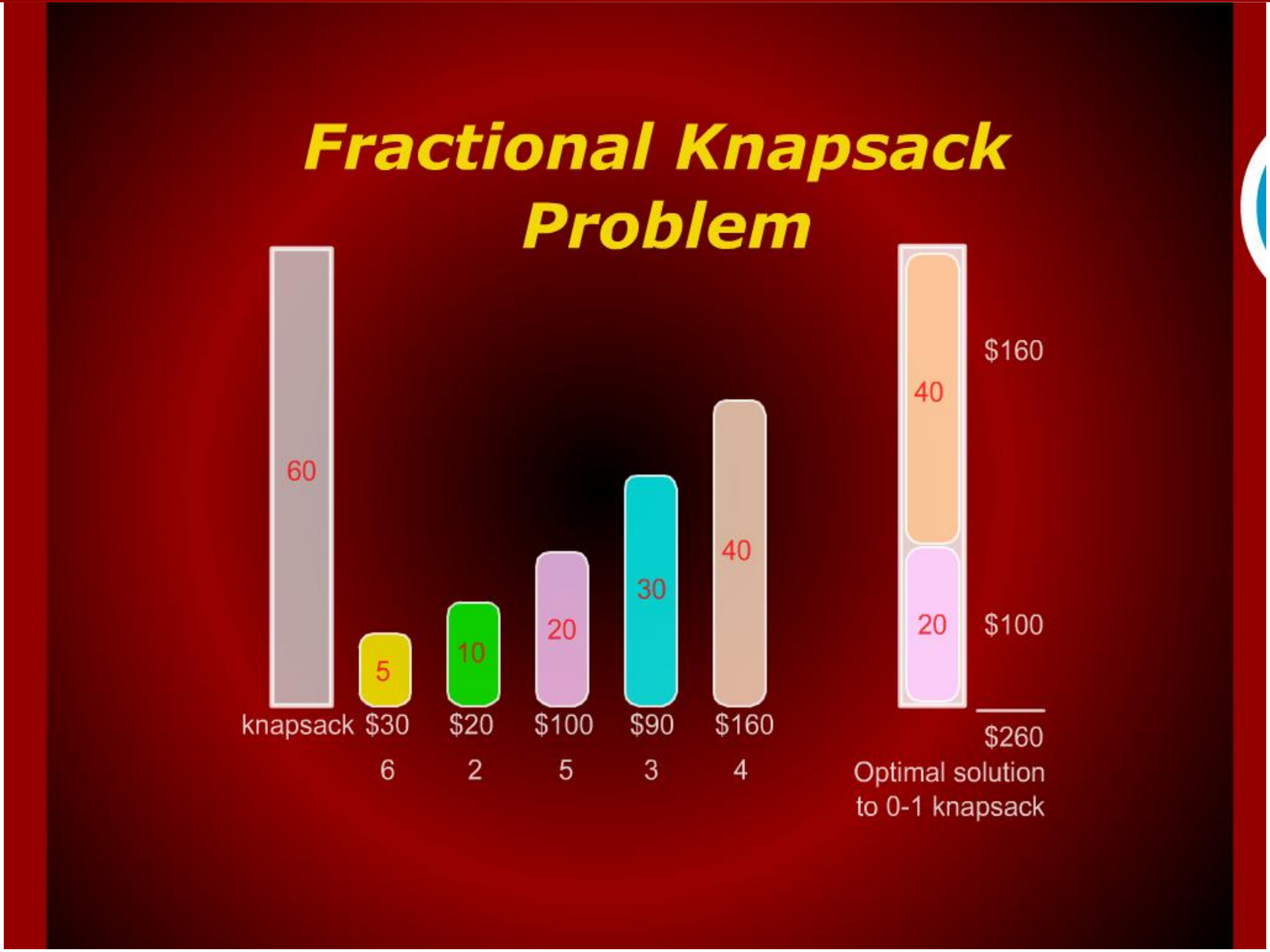
Degree of a Vertex

- In a digraph, the number of edges coming out of a vertex is called the out-degree of that vertex.
- Number of edges coming in is the in-degree.
- In an undirected graph, we just talk of degree of a vertex. It is the number of edges incident on the vertex.

Adjacent Vertices



Fractional Knapsack Problem



Fractional Knapsack

Fractional Knapsack Problem

Fractional Knapsack Problem

- Earlier we saw the 0-1 knapsack problem.
- A knapsack can only carry W total 3 weight.
- There are n items; the i^{th} item is worth v_i and weighs w_i.
- Items can either be put in the knapsack or not.

Course outline

Criteria for analyzing Algorithms

Criteria for analyzing Algorithms

Model of Computation

Model of Computation

Model of Computation

Random Access Machine